



Geospatial Modeling for Electrification Planning in the Democratic Republic of the Congo Using OnSSET



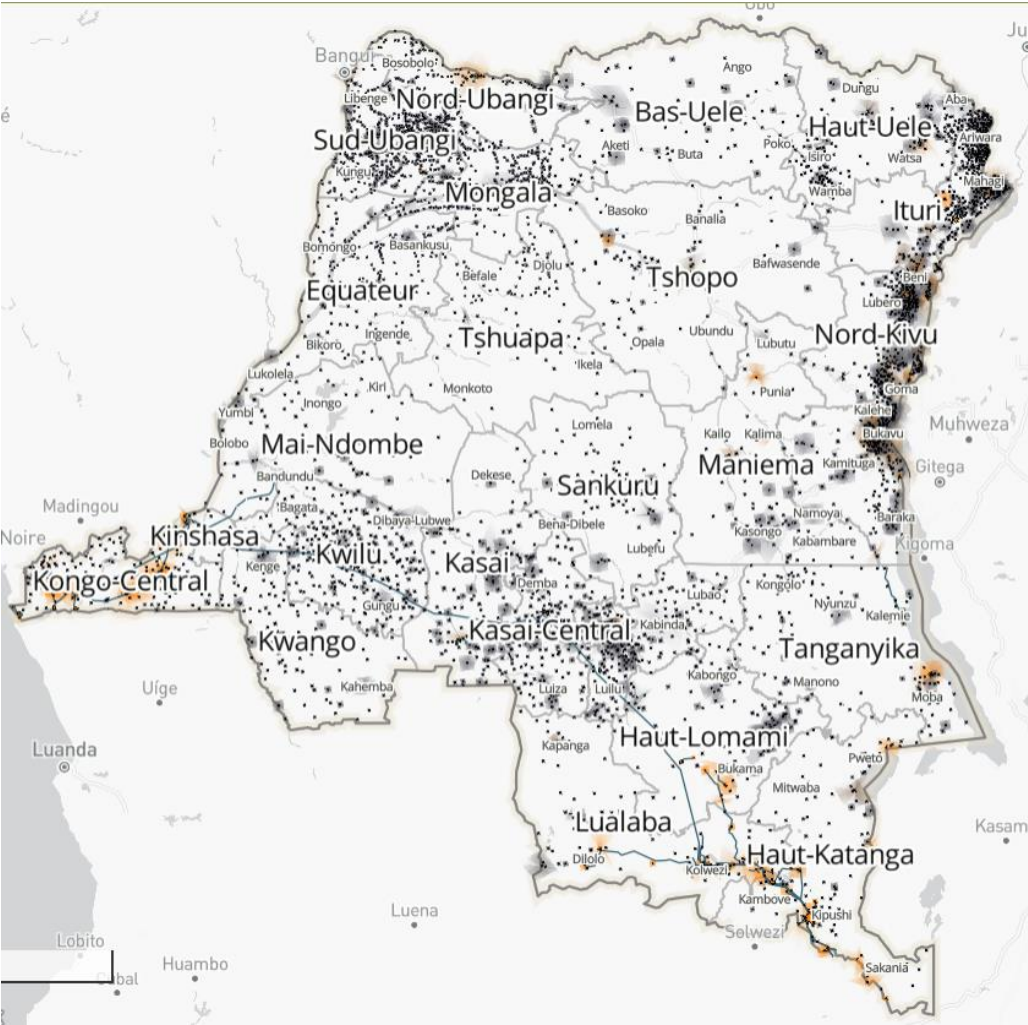
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Energy Modelling Platform Africa (EMP-A)

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Background

Aspect (2024)	Key Points
Population	127,4 millions
Superficies	2 345 410 km²
Average population density	36 people / km²
Energy potential	Hight renewable potential (Hydropower = 100 GW, Solar > 70 GW, Wind =15 GW, biomass, Geothermal, gas and uranium)
Generation capacity	3.64 GW
Electricity access	21% national access (44 % urban, 1% rural)
Main challenges	Weak infrastructure, financing gaps, institutional constraints, lack of energy planning
Government response	PDL 145 Territories, ANSER program, electricity market liberalization
2030 Target	62 % electricity access
International initiative	Joined World Bank Mission 300 (April 2024) to expand access, diversify energy, boost growth, jobs< and grid reliability.



Installed power per province

Objectives

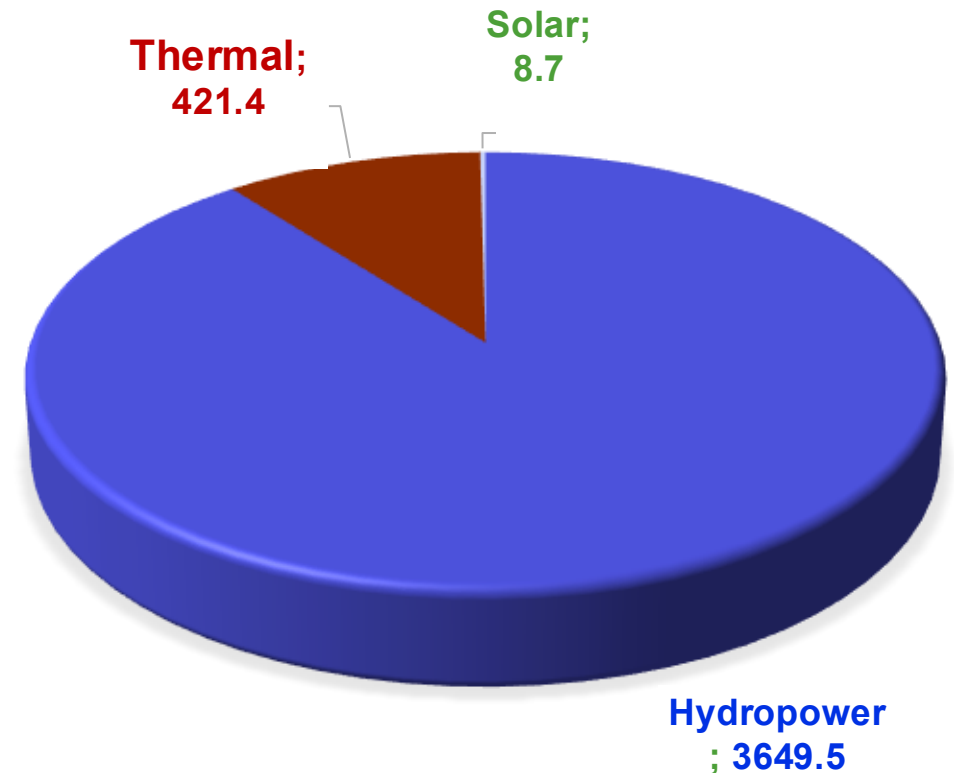
This study aims to provide technical and financial strategic results. The analyze will provide a spatiotemporal mapping of least-cost electrification, allowing for the identification of the most suitable technology (Solar Mini-grid-Hybrid, Minigrid-hydro, SHS, Wind and Grid extension) for the DRC, as well as the balance between the optimal investment and low emissions.

Research questions:

The main questions are :

- How can the proposed model effectively support the DRC in achieving its national energy compact plan's target of 62% electricity access rate by 2030?
- How the model can reduce the financing gap ?

INSTALLED CAPACITY (MW)



Source: Electricity Sector Regulatory Authority (ARE),
«Annual Report 2024 », 2025).

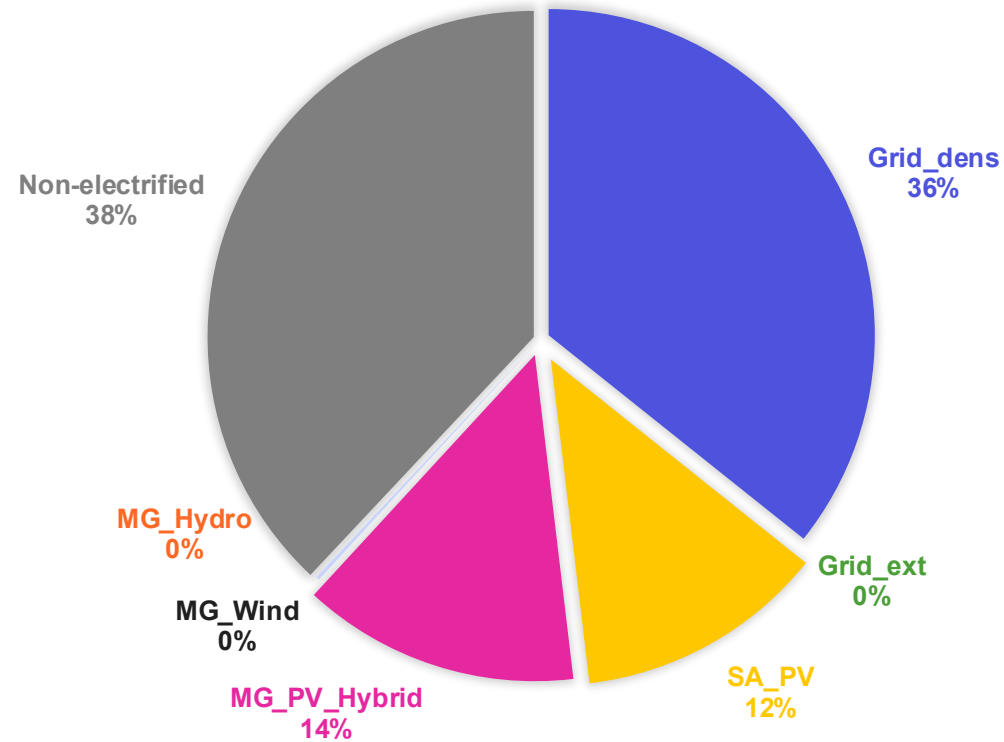
Scenarios

Using the OnSSET GIS Energy System Model the following scenarios were investigated. As the access target is less than 100 %, the strategy is to electrify the largest settlement first.

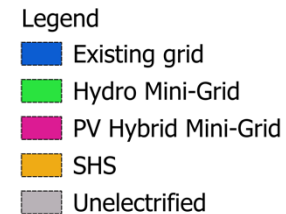
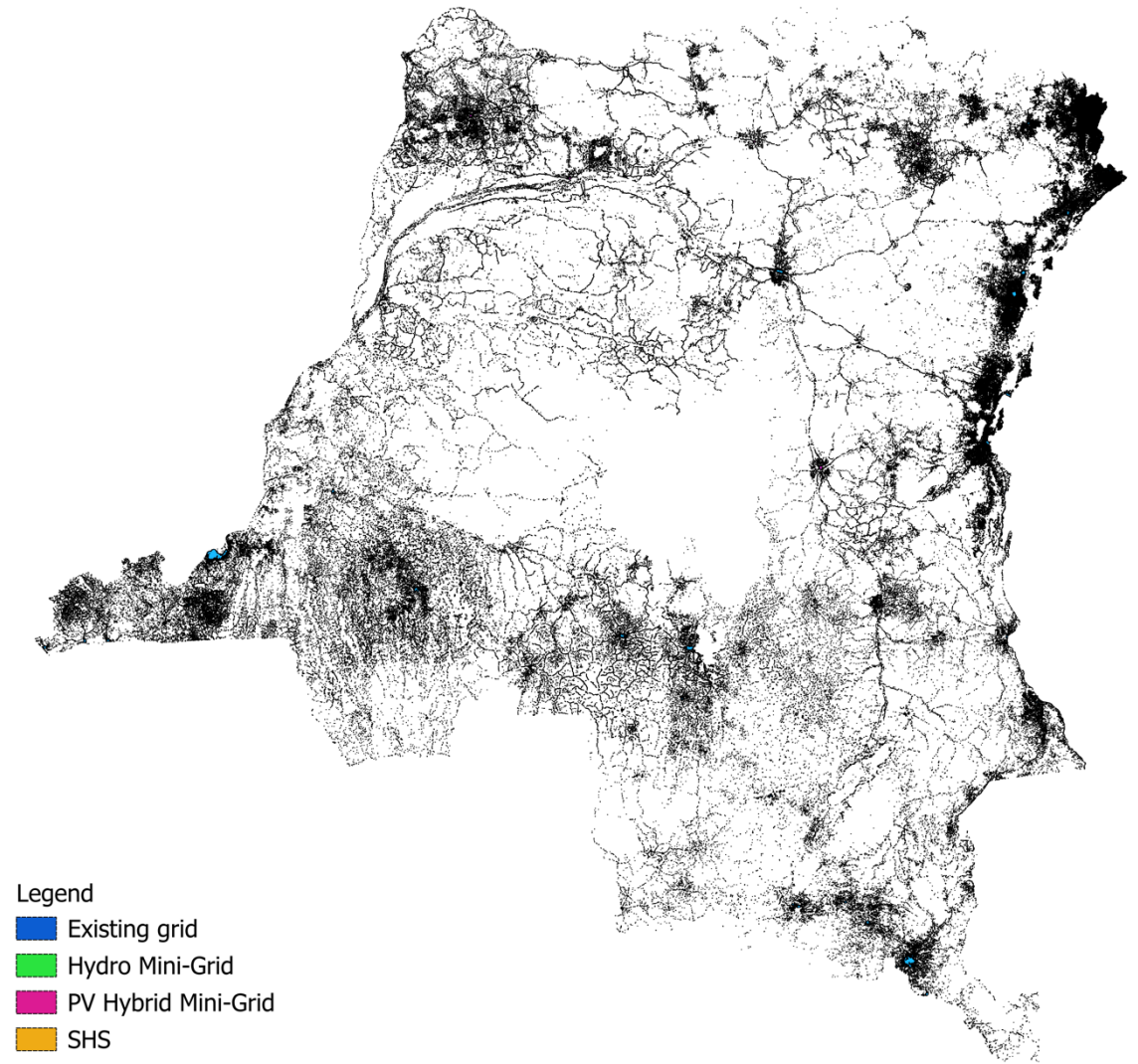
Scenario Label	Scenario Description	Key Assumptions
LCOE	This scenario aims to identify the most cost-effective path to electrification. It prioritizes cost reduction to achieve energy targets without imposing environmental restrictions.	• Start year electrification rate (2024): 21 % • Access rate Target (2030) : 62% • Urban and Rural tier : 6 • Max diesel : 100 % • Annual new grid connection : 60 000
Force grid	This scenario prioritizes the grid densification and grid extension without imposing environmental restrictions.	• Start year electrification rate (2024): 21 % • Access rate Target (2030) : 62% • Urban and Rural tier : 6 • Max diesel : 100 % • Annual new grid connection : no gap

Results of the LCOE scenario

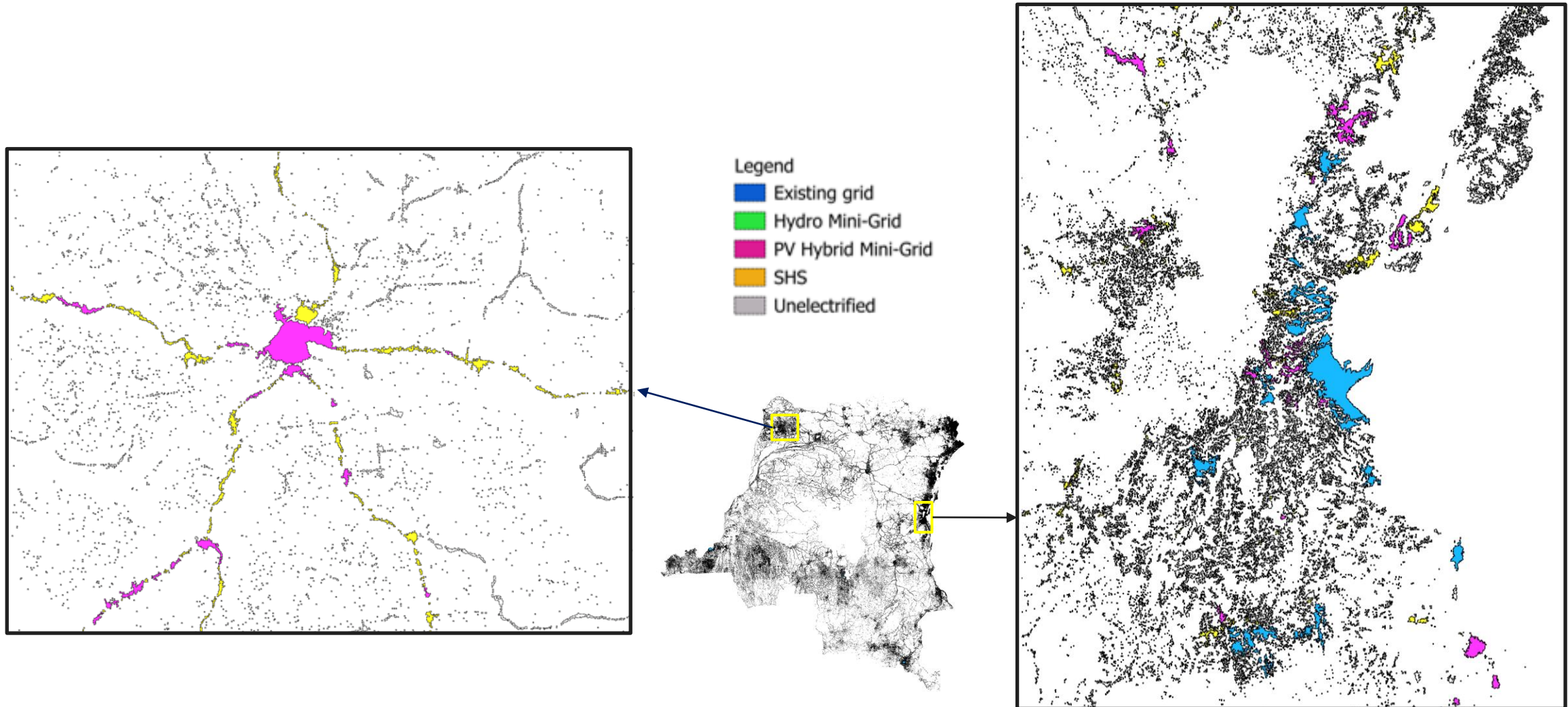
POPULATION 2030



This scenario prioritizes electrification through grid densification, followed by PV hybrid mini-grids and PV stand alone.

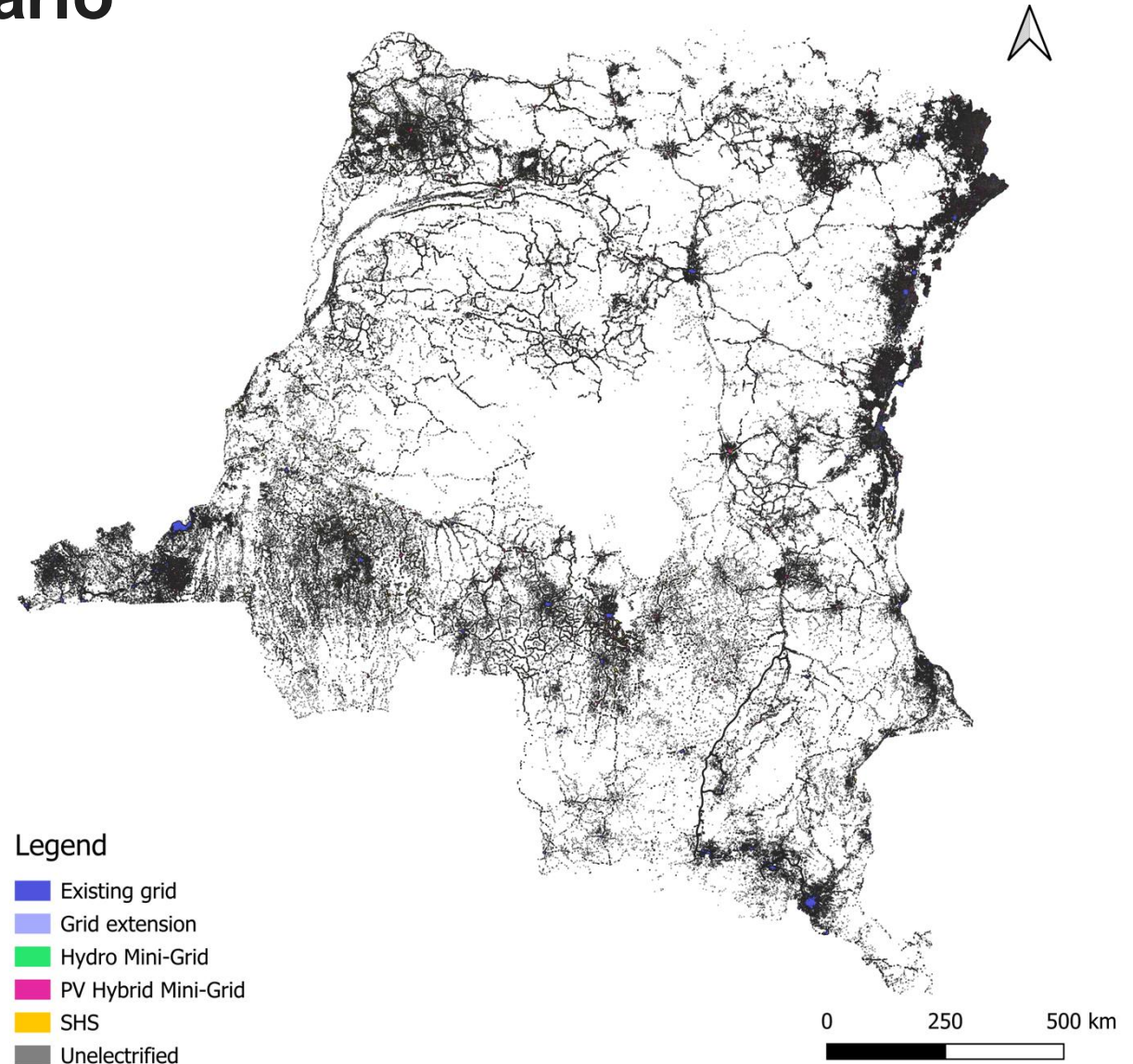
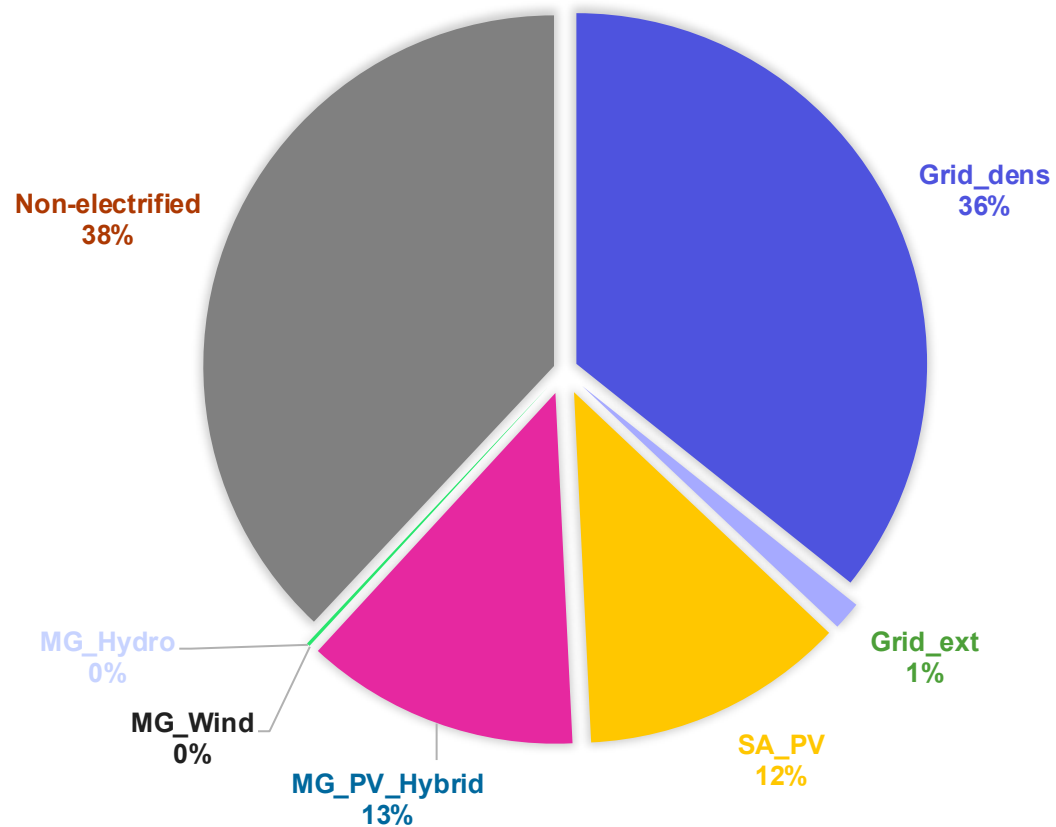


Results of the LCOE scenario

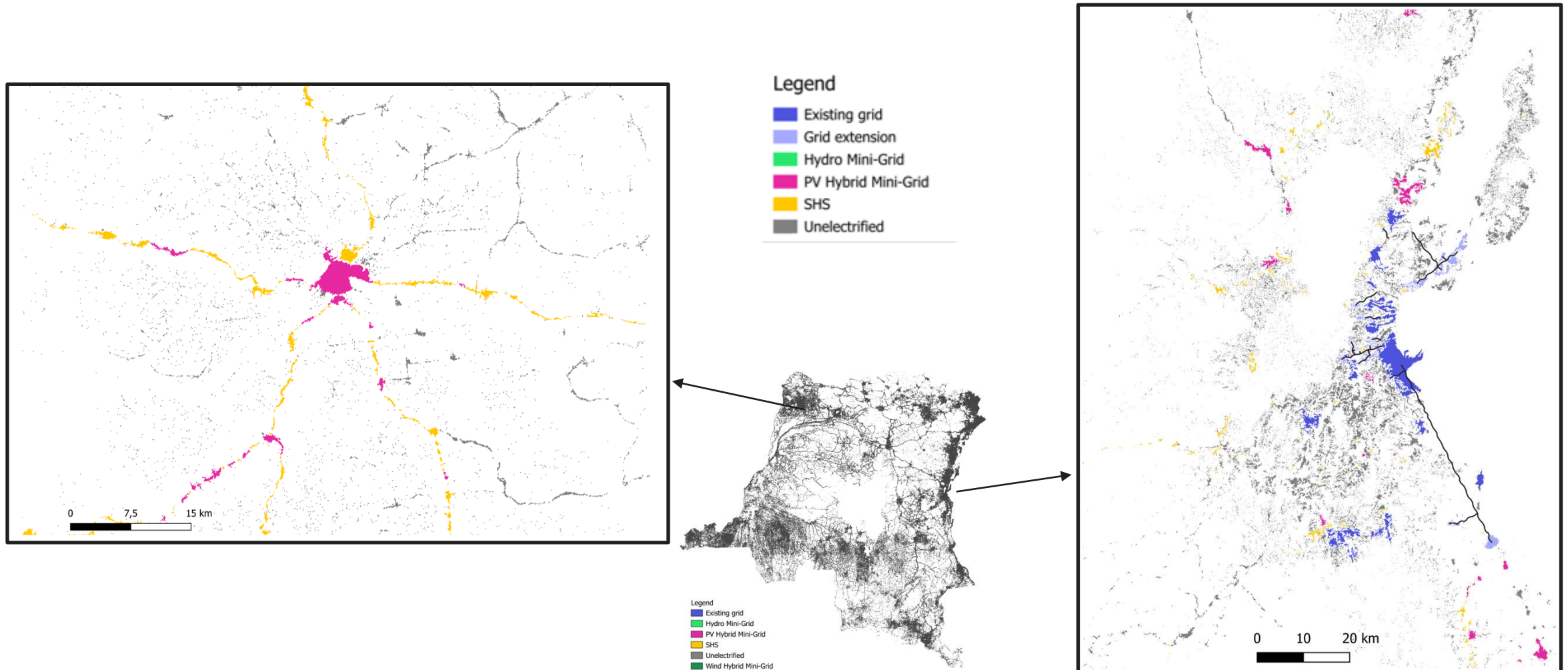


Results of the Force Grid scenario

POPULATION 2030

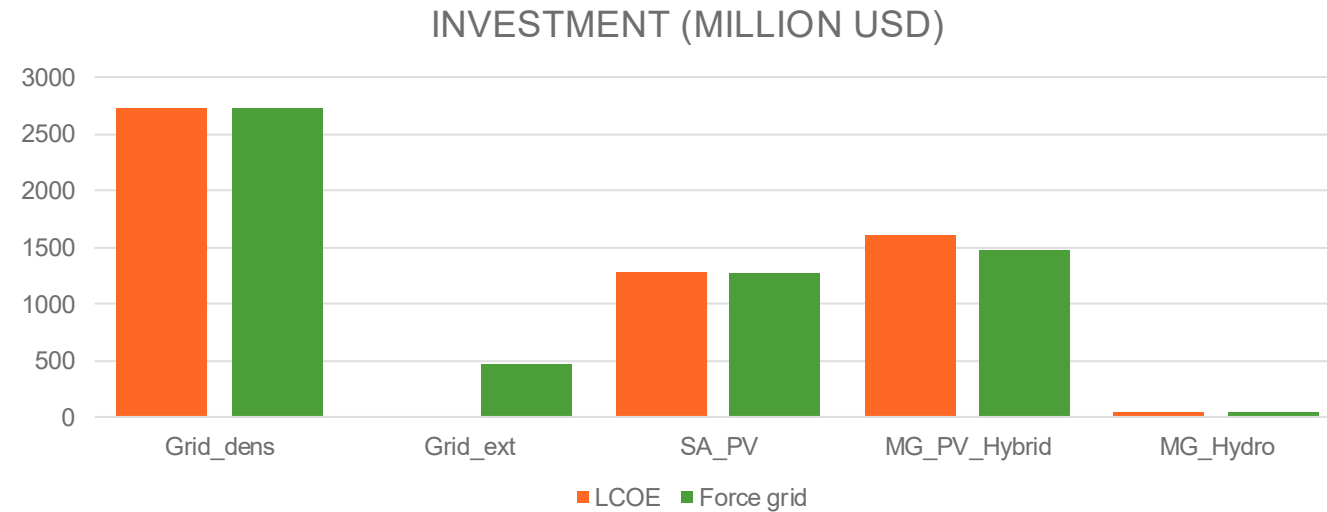


Results of the Force Grid scenario



Comparison of scenarios

Parameters	LCOE scenario	Force grid scenario
New connections	9 366 052	9 366 052
Investment (millions USD)	5636.42	5943.38
Emissions (tCO2)	2369.36	2216.17
Capacity (MW)	1376.29	1347.57



Conclusions and Policy Insights

To achieve the goal of 62% electricity access by 2030 at least cost according to the gap financing analysis, the government should prioritize the following investments:

- **Grid densification:** Invest \$2.7 billion to electrify large settlements within a maximum of 5 km from the existing grid.
- **Mini-grid hybrid PV-diesel:** Invest \$1.6 billion in systems with a total installed capacity of 423.27 MW, targeted at large settlements with high demand.
- **Solar home systems (SHS):** Invest \$1.28 billion to distribute kits in low-density, low-demand regions.
- **Mini-grid hydro:** Invest a smaller \$38.4 million in systems with 6.95 MW total capacity, deployed where viable hydro sites are nearby.

Future Work

- Onsset's results can be improved with a more realistic representation of the grid, taking into account its dynamic behavior and the technical constraints.
- Integrating AI could help bridge the data gap through learning from real-world projects.

THANKS